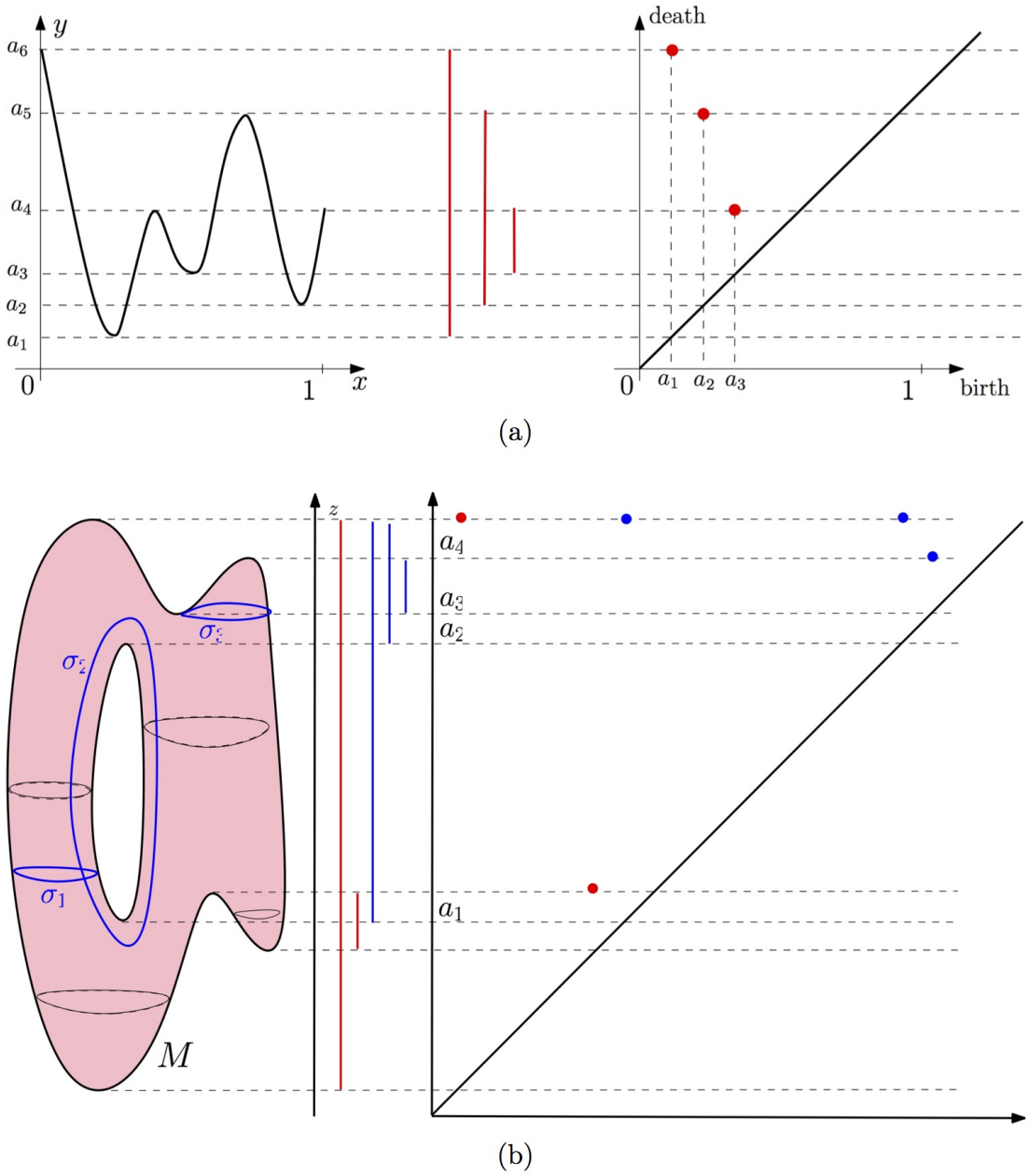

Figure 9: (a) Example 1: The persistence barcode and the persistence diagram of a function $f : [0, 1] \to \mathbb{R}$. (b) Example 2: the persistence barcode and the persistence diagram of the height function (projection on the z-axis) defined on a surface in $\mathbb{R}^3$.

features. When r goes through the height a_1, the sublevel sets F_r that were homeomorphic to two discs become homeomorphic to the disjoint union of a disc and an annulus, creating a first cycle homologous to σ_1 on Figure 9(b). A interval (in blue) representing the birth of this new 1-cycle is thus started at a_1. Similarly, when r goes through the height a_2 a second cycle, homologous to σ_2 is created, giving rise to the start of a new persistent interval. These two created cycles are never filled (indeed they span $H_1(M)$) and the corresponding intervals remains until the end of the filtration. When r reaches a_3, a new cycle is created that is filled and thus dies at a_4, giving rise to the persistence interval (a_3, a_4). So, now, the sublevel set filtration of f gives rise to two barcodes, one for 0-dimensional homology (in red) and one for 1-dimensional homology (in blue). As previously, these two barcodes can equivalently be represented as diagrams in the plane.

Example 3. In this last example we consider the filtration given by a union of growing balls centered on the finite set of points C in Figure 10. Notice that this is the sublevel set filtration of the distance function to C, and thanks to the Nerve Theorem, this filtration is homotopy equivalent to the Čech filtration built on top of C. Figure 10 shows several level sets of the filtration:
a) For the radius $r = 0$, the union of balls is reduced to the initial finite set of point, each of them corresponding to a 0-dimensional feature, i.e. a connected component; an interval is created for the *birth* for each of these features at $r = 0$.
b) Some of the balls started to overlap resulting in the *death* of some connected components that get merged together; the persistence diagram keeps track of these deaths, putting an end point to the corresponding intervals as they disappear.
c) New components have merged giving rise to a single connected component and, so, all the intervals associated to a 0-dimensional feature have been ended, except the one corresponding to the remaining components; two new 1-dimensional features, have appeared resulting in two new intervals (in blue) starting at their birth scale.
d) One of the two 1-dimensional cycles has been filled, resulting in its death in the filtration and the end of the corresponding blue interval.
e) all the 1-dimensional features have died, it only remains the long (and never dying) red interval. As in the previous examples, the final barcode can also be equivalently represented as a persistence diagram where every interval (a, b) is represented by the the point of coordinate (a, b) in $\mathbb{R}^2$. Intuitively the longer is an interval in the barcode or, equivalently the farther from the diagonal is the corresponding point in the diagram, the more persistent, and thus relevant, is the corresponding homological feature across the filtration. Notice also that for a given radius r, the k-th Betti number of the corresponding union of balls is equal of the number of persistence intervals corresponding to k-dimensional homological features and containing r. So, the persistence diagram can be seen as a multiscale topological signature encoding the homology of the union of balls for all radii as well as its evolution across the values of r.

5.3 Persistent modules and persistence diagrams

Persistent diagrams can be formally and rigorously defined in a purely algebraic way. This requires some care and we only give here the basic necessary notions, leaving aside technical subtelties and difficulties. We refer the readers interested in a detailed exposition to Chazal et al. (2016a).

Let Filt $= (F_r)_{r \in T}$ be a filtration of a simplicial complex or a topological space. Given a non negative integer k and considering the homology groups $H_k(F_r)$ we obtain a sequence of vector spaces where the inclusions $F_r \subset F_{r'}$, $r \leq r'$ induce linear maps between $H_k(F_r)$ and $H_k(F_{r'})$. Such a sequence of vector spaces together with the linear maps connecting them is called a *persistence module*.

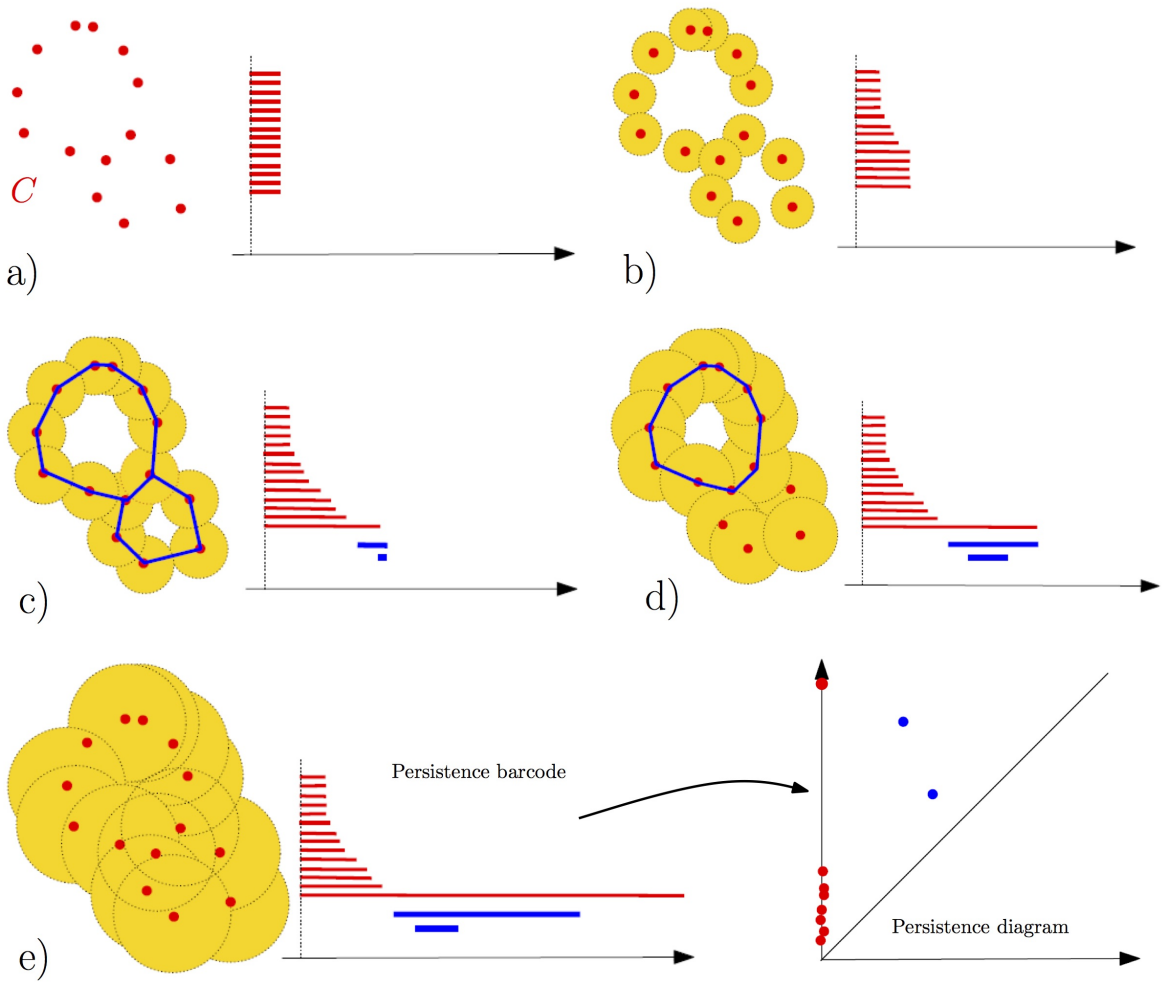

Figure 10: The sublevel set filtration of the distance function to a point cloud and the "construction" of its persistence barcode as the radius of balls increases. The blue curves in the unions of balls represent 1-cycles associated to the blue bars in the barcodes.